Skills Summary

Scatter Plots and Informal Lines of Fit

Use this reference while completing your worksheet or when studying.

Skill 1 — Reading a Point off a Scatter Plot

Every dot is one individual measured twice: its horizontal position is the first measurement, its vertical position is the second.

To read a value: find the given number on the horizontal axis, go straight UP to the dot, then straight ACROSS to the vertical axis. Report the vertical number — the horizontal one is what you were given.

Example: The points are $(1,4)$, $(2,7)$, $(3,9)$, $(4,12)$, $(5,14)$. What is the value when $x=3$?
Step 1. The question gives a horizontal value and asks for the matching vertical one, so find the dot above that number and read its height.
Step 2. Above $x=3$ the dot sits at height 9.
$\Rightarrow$ **9**

Try it: Points $(2,30)$, $(4,26)$, $(6,21)$, $(8,18)$. What is the value when $x=6$?

check: **21**

Skill 2 — Describing the Association

Direction: as x increases, does y tend to rise (POSITIVE), fall (NEGATIVE), or neither (NO ASSOCIATION)?

Strength: tightly packed along one path is STRONG; a wide fuzzy band is WEAK.

Shape: one straight path is LINEAR; a bend is NONLINEAR.

Then look for the exceptions: an OUTLIER sitting far from the rest, or CLUSTERS in separate groups. Always say direction, strength and shape — three words, not one.

Example: Describe $(1,3)$, $(2,5)$, $(3,8)$, $(4,10)$, $(5,14)$, and give the largest and smallest values.
Step 1. Read the vertical values in order of the horizontal ones: their trend gives the direction, and how tightly they follow it gives the strength.
Step 2. They climb steadily with no bend and no stray point: positive, strong, linear. Largest and smallest heights:
14 and **3**

Try it: Describe $(1,18)$, $(2,15)$, $(3,13)$, $(4,9)$, $(5,6)$, and give the largest and smallest values.

check: negative, strong, linear; **18** and **6**

Skill 3 — Drawing a Line of Fit and Writing It

Drawing it: lay a straight edge through the middle of the cloud so that roughly as many points sit above the line as below, and the points that miss it miss it by as little as possible. It does NOT have to pass through any data point, and it is never a line joining the first and last dots.

Writing it: pick TWO points that lie ON YOUR LINE. Then $m = \dfrac{y_2 - y_1}{x_2 - x_1}$, and put one of those points into $y = mx + b$ to get b.

Example: A line of fit passes through $(2, 5)$ and $(8, 17)$. Write its equation.
Step 1. Two points on the line are enough to pin it down: the slope comes from the pair, and the intercept from putting one point back into the equation.
Step 2. $m = \dfrac{17 - 5}{8 - 2} = 2$; then $5 = 2(2) + b$ gives $b = 5 - 4$.
$m = \mathbf{2}$, $b = \mathbf{1}$, so $y = 2x + 1$.

Try it: A line of fit passes through $(1, 20)$ and $(6, 10)$. Find m and b.

check: $m = \mathbf{-2}$, $b = \mathbf{22}$

Skill 4 — Predicting from the Line

To predict: put the given x into $y = mx + b$ and work out y. Write the substitution before the answer.

Say which kind of prediction it is: INTERPOLATION inside the range of the data (trustworthy) or EXTRAPOLATION outside it (risky, and worthless once it gives an impossible value).

Reading the slope: m is how much y changes for each 1 that x increases — an association in this sample, not proof of a cause.

Example: Use $y = 2x + 1$ to predict the value at $x = 12$.
Step 1. A prediction from a line of fit is just a substitution, so replace the letter with the given number and evaluate.
Step 2. $y = 2(12) + 1 = 24 + 1$.
$\Rightarrow$ **25**

Try it: Use $y = -2x + 22$ to predict the value at $x = 9$.

check: **4**